Metallicrab - Final Results

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Introduction

- The purple shore crab, *Hemigrapsus nudus*, is a
  - common intertidal species located in the Pacific Northwest that lives in nearshore habitats
  - often where metals are introduced into the ocean from urban runoff, sewage, and shipping ports
- *H. nudus* are osmoregulators
  - previous studies have shown that their hemolymph osmolality stays relatively stable across exposure to a broad range of salinities, indicating that their internal water and ion balance is closely tied to environmental stress (Adeleke et al. 2021; Lignot et al. 2000).
- Copper exposure in crustaceans is known to cause multiple physiological disruptions, including oxidative stress and altered immune responses (Truscott and White 1990).
  - Although copper is an essential trace element and a component of hemocyanin (hemolymph), excessive concentrations can still be toxic (Coates and Costa-Paiva 2020).
- Lead is a nonessential metal known to affect reproductive and oxidative mechanisms, and has a stronger effect in weak osmoregulators than osmoconformers (Li et al. 2015; Amado et al. 2012; Rainbow 2002).

Background

- The Washington State Department of Health states that lead exposure can cause long-term harm and is especially toxic to children, with studies linking lead exposure to behavioral challenges.
  - In Washington state alone, more than 2,500 children under the age of 6 tested between 2019 and 2023 had blood lead levels at or above 5 µg/dL (Washington State Department of Health).
- Copper and lead are of concern because they can accumulate in aquatic and terrestrial organisms and impact essential physiological processes, even at non-lethal concentrations (Amado et al. 2012; Rainbow 2002).
  - Researchers found that the survival rate of sperm decreased after lead exposure in freshwater crabs as the accumulation of lead concentrated in their bodies was statistically significant (Li et al. 2015).
  - In another study, researchers found that stress often results in a decrease in ionic regulation originating in cellular alterations of Na+/K+-ATPase activity (Lignot et al. 2000).
  - Heavy metals also affect hemocytes, the cells that manage innate immunity. Changes in total hemocyte count (THC) may reflect either immune activation or suppression, depending on the dose and length of exposure (Yildiz and Atar 2002).

Research Questions

- How does the presence of heavy metals, specifically copper and lead, affect *Hemigrapsus nudus* physiology?
- Does copper and lead exposure alter
  - Osmolality
  - Righting time
  - Hemocyte count
- What do these responses suggest?
- Hypotheses:
  - H0: There will be no change in hemocyte count across all groups.
  - HA: There will be an impact on the total hemocyte count (THC) in the trace metal experimental groups.

  - HA1: THC will increase (immune activation)
  - HA2: THC will decrease (immune suppression)

- H0: There will be no change in righting time across all groups.
- HA: Increase in righting time (stress response) in the trace metal experimental groups.

- H0: There will be no change in osmolality across all groups
- HA: There will be an increase in osmolality in the trace metal experimental groups

Experimental Design + Methods

Week 1

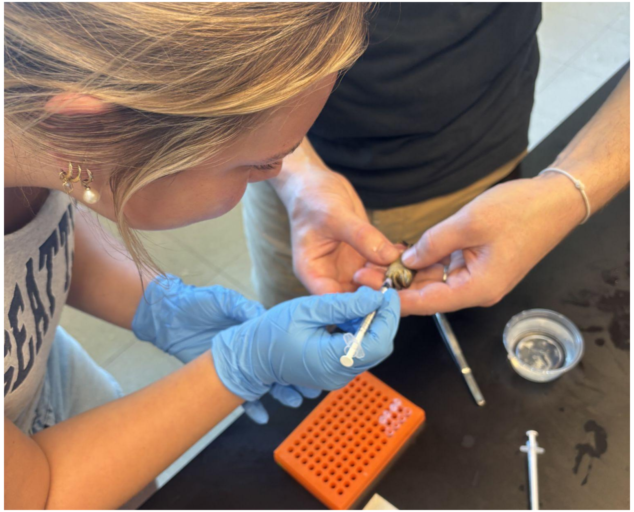
- 14 similarly sized *H. nudus* randomly selected
- 2 crabs used for initial hemolymph sampling
- Remaining 12 crabs tagged and assigned to:
    - Control treatment
    - Copper treatment
    - Lead treatment
- 4 crabs per treatment group
- Recorded righting time for all crabs
- Treatments were kept at 35 ppt and the same temperature
- Added EDTA anticoagulant to extractions

Week 2

- Extracted hemolymph from 2 crabs per treatment
- Recorded righting time
- Added EDTA anticoagulant to extractions

Week 3

- Extracted hemolymph from all remaining crabs
- Recorded righting time
- Used hemocytometer for hemocyte counts
- Used osmometer for osmolality measurements on all samples
- Data were compared across treatments using graphs and ANOVA.



Results - Righting Time

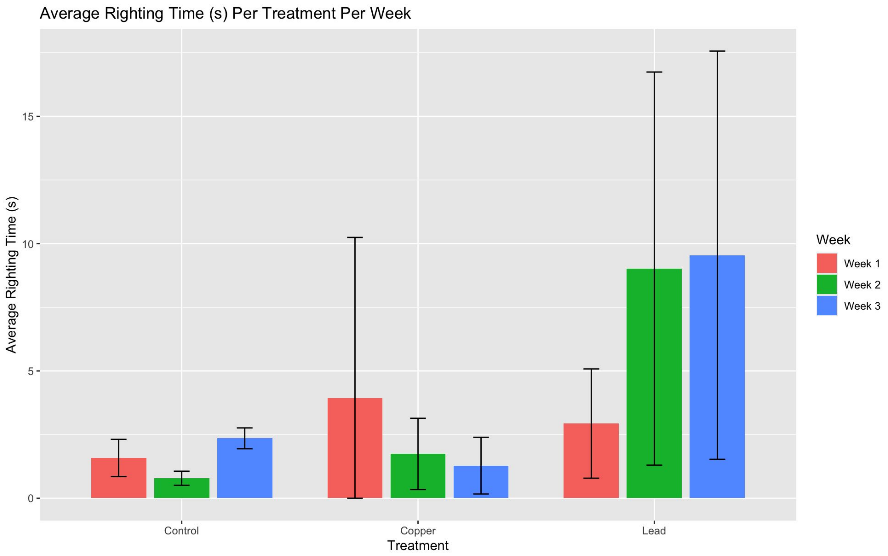


Figure 1. Average righting time (s) per treatment (control, copper, lead) over three weeks. Bar graph with standard deviation error bars. Fastest average righting time found in control treatment. Variation in copper and lead treatments compared to control with lead having the slowest average righting time over the three weeks.

- One-way ANOVA: Average righting times per treatment
  - $p = 0.0649$

Increase in righting time showed in lead treatment but, results not statistically significant

*Heavy metals affecting righting time may show with more replicates

Results - Hemocyte Count

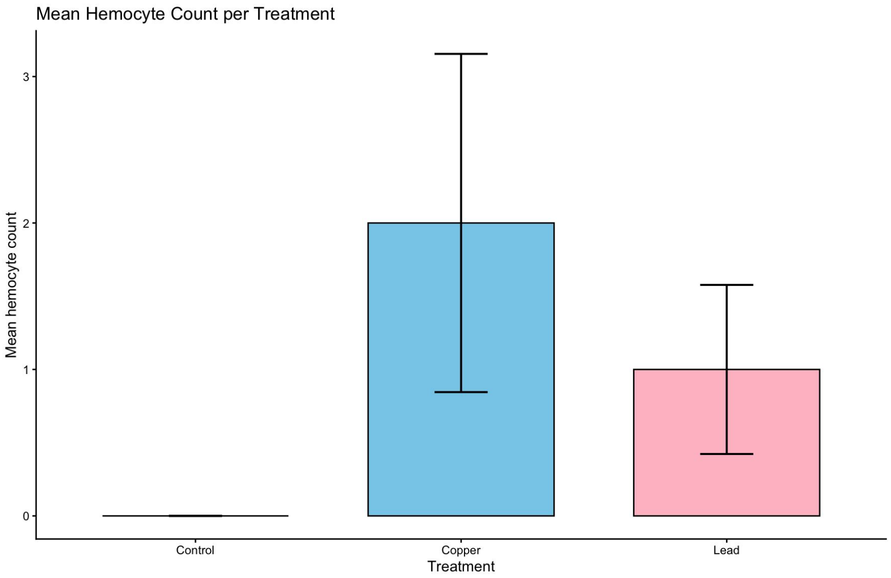


Figure 2. Mean hemocyte count per treatment. Bar graph with standard error bars of 95% confidence interval. Found no hemocytes in control treatment. Increased hemocyte count in both copper and lead treatments with copper having the highest mean count.

- One-way ANOVA: Effect of heavy metals on total hemocyte count per treatment
- $P = 0.244$

Small sample size of n=9 as centrifuged hemolymph samples broke cells

*Increase in hemocytes may suggest immunological stress response, likely increased to fight potential invader in their bodies

Results - Osmolality

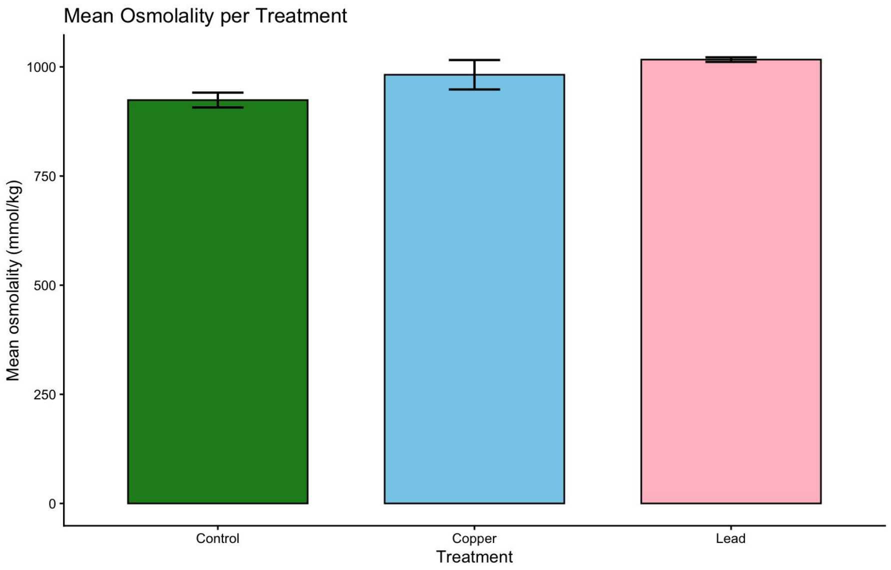


Figure 3. Mean osmolality (mmol/kg) per treatment. Bar graph with standard error bars of 95% confidence interval. Control treatment shows the lowest mean osmolality. Copper and lead treatments show increased osmolality compared to control treatment with lead having the highest mean values.

- One-way ANOVA: Effect of heavy metals on mean osmolality of each treatment
- P = 0.0633
- Second one-way ANOVA: Mean osmolality of lead treatment compared to control
- P = 0.0065
- Osmolality refers to the concentration of dissolved solutes in the hemolymph

*Statistically significant result suggests that lead treatment crabs are more physiologically stress compared to control crabs

Conclusion-Maria

- Heavy metal exposure caused measurable physiological stress in Hemigrapsus nudus.
- Lead exposure was associated with longer righting times, suggesting impaired behavioral performance.
- Copper exposure resulted in higher osmolality values, indicating possible disruption of osmoregulation.

Future Work- Maria

- Use a larger sample size to make the results more reliable.
- Use exact metal concentrations instead of metal weights.
- Test different metal concentrations, like low, medium and high doses.
- Monitor crabs for a longer time to see long term effects.
- Use more consistent hemolymph collection methods to avoid damaging cells.

References

Adeleke, B., Robertson-Andersson, D., & Moodley, G. (2021). Osmotic response of *Dotilla fenestrata* (sand bubbler crab) exposed to combined water acidity and varying metal (Cd and Pb). *Heliyon*, *7*(4), e06763. https://doi.org/10.1016/j.heliyon.2021.e06763

Amado, E. M., Freire, C. A., Grassi, M. T., & Souza, M. M. (2012). Lead hampers gill cell volume regulation in marine crabs: stronger effect in a weak osmoregulator than in an osmoconformer. *Aquatic toxicology (Amsterdam, Netherlands)*, *106-107*, 95–103. https://doi.org/10.1016/j.aquatox.2011.10.012

Coates, C. J., & Costa-Paiva, E. M. (2020). Multifunctional Roles of Hemocyanins. *Sub-cellular biochemistry*, *94*, 233–250. https://doi.org/10.1007/978-3-030-41769-7_9

EPH-WTN--Washington Tracking Network--4300. (n.d.). *Lead risk and exposure data*. Washington State Department of Health. https://doh.wa.gov/data-and-statistical-reports/washington-tracking-network-wtn/lead-risk-and-exposure

Li, N., Hou, Y., Ma, D., Jing, W., Dahms, H., & Wang, L. (2015). Lead accumulation, oxidative damage and histopathological alteration in testes and accessory glands of freshwater crab, Sinopotamon henanense, induced by acute lead exposure. *Ecotoxicology and Environmental Safety*, *117*, 20–27. https://doi.org/10.1016/j.ecoenv.2015.03.019

Rainbow, P. S. (2002). Trace metal concentrations in aquatic invertebrates: why and so what? *Environmental Pollution*, *120*(3), 497–507. https://doi.org/10.1016/s0269-7491(02)00238-5

Truscott, R., & White, K. N. (1990). The Influence of Metal and Temperature Stress on the Immune System of Crabs. *Functional Ecology*, *4*(3), 455–461. https://doi.org/10.2307/2389609

YILDIZ, H. Y, & ATAR, H. H (2002). Haemocyte Classification and Differential Counts in the Freshwater Crab, Potamon fluviatilis. *Turkish Journal of Veterinary & Animal Sciences 26* (2): 403-406.